MIL-S-48305(PA)
15 September 1976
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

SWITCH, SQUIB ACTUATED: M60, 61, 63, 64, 65, 66, 67, 68, 73

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for four types of six-pole squib actuated switches, delay and non-delay, entitled Switch, Squib Actuated: M60, 61, 62, 63, 64, 65, 66, 67, 68, and M73 which are used in the Lance warhead section.

1.2 Classification. This specification covers switches of the following:

Type I	Non-delay.
Type II	0.1 second delay.
Type III	1.0 second delay.
Type IV	2.0 second delay.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	-	Environmental Testing, Aeronautical and Associated Equipment, General Specification For.
MIL-STD-1460	-	Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

FSC 1135

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-271	-	Nondestructive Testing Requirements for Metals.
MIL-STD-331	-	Temperature and Humidity Test for Use in Development of Fuzes.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

DRAWINGS

MILITARY

8862872	-	Switch, Squib Actuated Non-Delay, M60.
8862873	-	Switch, Squib Actuated Delay, M67.
8862878	-	Switch, Squib Actuated Non-Delay, M61.
8878974	-	Switch, Squib Actuated Delay, M68.
8878975	-	Switch, Squib Actuated Delay, M65.
8878976	-	Switch, Squib Actuated Non-Delay, M62.
8878978	-	Switch, Squib Actuated Non-Delay, M63.
8878979	-	Switch, Squib Actuated Delay, M64.
8878980	-	Switch, Squib Actuated Delay, M66.
8862880	-	Switch, Squib Actuated Delay, M73.

PUBLICATIONS

PICATINNY ARSENAL

EL-8862872	-	Switch, Squib Actuated Non-Delay, M60.
EL-8862873	-	Switch, Squib Actuated Delay, M67.
EL-8862878	-	Switch, Squib Actuated Non-Delay, M61.
EL-8878974	-	Switch, Squib Actuated Delay, M68.
EL-8878975	-	Switch, Squib Actuated Delay, M65.
EL-8878976	-	Switch, Squib Actuated Non-Delay, M62.
EL-8878978	-	Switch, Squib Actuated Non-Delay, M63.
EL-8878979	-	Switch, Squib Actuated Delay, M64.
EL-8878980	-	Switch, Squib Actuated Delay, M66.
EL-8862880	-	Switch, Squib Actuated Delay, M73.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

3.2 Assembly. The switch shall comply with all requirements specified on the drawings (dwgs) 8862872, 8862873, 8862878, 8878974, 8878975, 8878976, 8878978, 8878979, 8878980, or 8862880 as applicable, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Seal integrity.

3.5.1 Gross leakage. The switch shall show no evidence of leakage when, after initial pressurization of the previously sealed assembly, the switch is immersed in two inches of water under a reduced atmospheric pressure of 5+1 pounds per square inch absolute. After testing as specified in 4.4.3.1, the switch shall be tested for compliance with 3.5.2.

3.5.2 Fine leakage. The switch shall demonstrate an indicated leakage rate no greater than 1×10^{-5} atmosphere cc/second when subjected to the helium leak test of 4.4.3.2, except that 2×10^{-5} atmosphere cc/second shall be considered an acceptable limit in the case of a repeat test for lot sample data.

3.6 Bridge resistance. The bridge resistance of the switch shall be 1.8 $\pm$ 0.2 ohms when tests as specified in 4.4.4.

3.7 Circuit resistance. The circuit resistance of all closed switch contact, pre-function and post-function, shall be 0.008 ohms maximum when tested as specified in 4.4.5.

3.8 Insulation resistance. The insulation resistance (IR) between each isolated test point and all other test points electrically tied together shall be a minimum of 50 megohms when tested as specified in 4.4.6. Said test points shall include, exclusively, the switch housing, shunted squib leads, and all individual switch terminals, except that closed terminal pairs shall constitute single test points. Stated IR requirement shall apply both before and after functioning, with the exception that squib terminals will not be included in post-function IR measurement.

3.9 Squib energy.

3.9.1 Non-function. The switch shall not function when the squib terminals are subjected to a continuous current of 100 +10-00 milliamperes direct current (dc) for a minimum of 10 seconds. Testing shall be as specified in 4.4.7.

3.9.2 Function. The switch shall function when the squib terminals are subjected to a continuous current of 1.00 +0.00 -0.05 amperes (dc). Functioning time for each switch type shall be within the limits specified in Table I. Said time shall be defined as commencing with the first application of firing current to the squib terminals and end with the closing of the last normally open switch contact or opening of the last normally closed switch contact, whichever

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

occurs later. After functioning, the switch shall be tested for compliance with the requirements for insulation resistance 3.8, circuit resistance 3.7, and radiographic examination post-function inspection 3.10.1.2; there shall be no evidence of ruptures in the metallic case and the external appearance shall be identical to that of the unfunctioned switch. Testing shall be as specified in 4.4.8.

Table I

M No.	Dwg No.	Normally Closed	Normally Open	Time	Functioning Time
M60	8862872	6,5,4,3,2,1	None	0.0 Sec. (Non-delay)	0.01 sec. max.
M61	8862878	6,5	4,3,2,1	0.0 Sec. (Non-delay)	0.01 sec. max.
M62	8878976	6,5,4	3,2,1	0.0 Sec. (Non-delay)	0.01 sec. max.
M63	8878978	6,5,4,3,2	1	0.0 Sec. (Non-delay)	0.01 sec. max.
M68	8878974	6,5,4,3	2,1	0.1 Sec. (Delay)	0.100 ± 0.040 Sec.
M64	8878979	6,5	4,3,2,1	0.1 Sec. (Delay)	0.100 ± 0.040 Sec.
M65	8878975	6,5,4	3,2,1	1.0 Sec. (Delay)	1.00 ± 0.25 Sec.
M67	8862873	6,5,4,3	2,1	2.0 Sec. (Delay)	2.00 ± 0.70 Sec.
M66	8878980	6,5,4	3,2,1	2.0 Sec. (Delay)	2.00 ± 0.70 Sec.
M73	8862880	6,5,4,3	2,1	1.0 Sec. (Delay)	1.00 ± 0.25 Sec.

Note: The normally closed switch contacts shown in Table I shall be interpreted to apply prior to switch functioning. After functioning, the contacts shall in each case be the reverse of that shown in Table I.

3.10 Environmental testing. The switch shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The switch shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.9. Tests shall be conducted in the order specified.

3.10.1 Radiographic examination.

3.10.1.1 Pre-function inspection. All switch assemblies shall be radiographically inspected by X-ray and Neutron radiography to assure against the following defects: improper solder placement, evidence of foreign material, cavity contains no charge, charge contains voids, improper position of normally closed switch contacts, and normally open switch contacts, malfunction of parts, component missing or improper, component damaged or loose, improper assembly of components, and cocked piston. Testing and inspection shall be in accordance with MIL-STD-271 and the best commercial neutron radiography procedures, where applicable, and as specified in 4.4.9.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

3.10.1.2 Post-function inspection. All switch assemblies subjected to destructive testing (see applicable sampling requirements) shall be only X-ray radiographically inspected to assure that individual switch contacts are free from deformation which is potentially detrimental to circuit continuity. Testing and inspection shall be in accordance with MIL-STD-271, where applicable, and as specified in 4.4.9.1.

3.10.2 Temperature shock. The switch shall withstand temperature cycling between +160 ± 5°F and -65 ± 5°F for three complete cycles without evidence of physical damage (see 6.5). Testing shall be as specified in 4.4.9.2.

3.10.3 Transportation vibration. The switch shall withstand vibration at either temperature extreme (+160 ± 5°F or -65 ± 5°F) in each of three mutually perpendicular axes when tested to the frequencies, amplitudes and durations specified below. There shall be no evidence of physical damage (see 6.5) to the switch. Testing shall be as specified in 4.4.9.3.

Frequency (Hz)	Acceleration Level or Double Amplitude	Cycle Time (Minutes)
5-10-5	0.8 inch	4 (2 up, 2 down)
10-15-10	0.4 inch	4 (2 up, 2 down)
15-80-15	5 g's	4 (2 up, 2 down)
80-115-80	0.015 inch	4 (2 up, 2 down)
115-500-115	10 gravities (g's)	4 (2 up, 2 down)

3.10.4 Mechanical shock. The switch shall show no evidence of physical damage (see 6.5) when subjected to mechanical shock. Shocks of 50 ± 5 (g's) peak amplitude, half-sine duration of 11 ± 1 milliseconds (ms) shall be applied once in each direction along each of the three principal axes for a total of 6 shocks. Testing shall be as specified in 4.4.9.4.

3.10.5 Random vibration. The switch shall withstand random vibrations that are applied along each of the three mutually perpendicular axes at the stresses and for the durations specified in 4.4.9.5. There shall be no evidence of physical damage (see 6.5).

3.10.6 Low temperature.

3.10.6.1 Soak. The switch shall withstand a temperature of -65 ± 5°F for a minimum of 72 hours without evidence of physical damage (see 6.5). Testing shall be as specified in 4.4.9.6.1.

3.10.6.2 Function. The switch shall be stabilized (see 6.4) at a temperature of -65 ± 5°F and, when tested in accordance with 4.4.9.6.2, the switch shall meet the requirements of squib energy function (3.9.2), except that insulation resistance (3.8) will not be determined. The requirement for radiographic, post-function inspection, (3.10.1.2) shall also apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

3.10.7 High temperature.

3.10.7.1 Soak. The switch shall withstand a temperature of +160 ± 5°F for a minimum of 72 hours without evidence of physical damage (see 6.5). Testing shall be as specified in 4.4.9.7.1.

3.10.7.2 Function. The switch shall be stabilized (see 6.4) at a temperature of +160 ± 5°F, and, when tested in accordance with 4.4.9.7.2, the switch shall meet the requirements of squib energy function (3.9.2). The requirement for radiographic, post-function, (3.10.1.2) shall also apply.

3.10.8 Temperature and humidity. The switch shall withstand one complete 14-day temperature and humidity cycle which shall consist of cycling the switch nine times between the extremes of +160°F (95 percent relative humidity rh) and -65°F without evidence of physical damage (see 6.5). Testing shall be as specified in 4.4.9.8.

3.11 Reliability control. To establish a measure for determining the margin of safety and quality of the product which characteristically effects unit reliability, variable type data shall be recorded for lot acceptability. A lot will be considered acceptable whenever failure(s) of the test parameters at high or low temperatures do not exceed the acceptance number(s) specified in Table II. Testing shall be as specified in 4.4.10.

Table II

Test	Acceptance Number: High Temperature +195 ± 5°F	Acceptance Number: Low Temperature -80 ± 5°F
a. Functioning Time (Table I)	1	1
b. Insulation resistance (3.8)	0	N/A
c. Circuit resistance (3.7)	0	0
d. Radiographic examination (3.10.1.2)	0	0

3.12 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460. In addition, prior to potting all soldered assemblies shall comply with the requirements of MIL-STD-1460.

3.13 First article approval. Approval of the contractors capability to produce the end item switch either before, or in the initial stage of the production contract involves testing and evaluating the first article sample for conformance with the specification requirements. This includes a determination of the probability of compliance (4.4.2) that is associated with the factorial test plan (4.4.1). The functioning time (see 3.9.2) will be the most severely

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

affected when the switch is functioned at high or low temperatures; therefore, a minimum "K" value of +2.22 is required for acceptance of each first article test except a minimum "K" value of +1.16 shall apply to the 0.1 second delay at the low temperature requirement only.

3.14 Workmanship.

3.14.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.14.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.14.3 Burrs. Parts shall be free from burrs.

3.14.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.14.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.14.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 16 complete assemblies (see dwgs specified in 3.2) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

Table III

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Seal integrity (see 3.5)		X
3. Bridge resistance (see 3.6)		X
4. Circuit resistance (see 3.7)		X
5. Insulation resistance (see 3.8)		X
6. Squib energy (see 3.9)	X	
7. Radiographic examination (see 3.10.1)		X
8. Temperature shock (see 3.10.2)		X
9. Transportation vibration (see 3.10.3)	X	
10. Mechanical shock (see 3.10.4)	X	
11. Random vibration (see 3.10.5)		X
12. Low temperature (see 3.10.6)		X
13. High temperature (see 3.10.7)		X
14. Temperature and humidity (see 3.10.8)	X	
15. First article approval (see 3.13)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. All items in each inspection lot shall be from one production line, one shift, one batch of products, one manufacturer of raw material or components, and one production method or manufacturing operation. Lot size shall be limited to 1,000 units maximum. A lot shall contain not more than one switch type of the four listed, (non-delay, 0.1 second delay, 1.0 second delay, or 2.0 second delay).

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 For various type squib actuated switches, (see drawings listed in 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Squib leads not shunted	Visual
	102. Incorrect location of terminal pins	Gage
	103. Case damaged	Visual
	104. Marking illegible, improper or missing	Visual
	105. Objective evidence of inspection for solder (see 3.12)	Visual
	106. Objective evidence relative to material missing or improper (see 3.1)	Visual
	107. Serialization missing or improper (see 3.4)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.14)	Visual

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

Major:

101. Preservation, packaging and packing not in accordance with MIL-STD-1169.....Visual

102. Marking not in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.14).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Seal integrity	Major
b. Bridge resistance	Major
c. Circuit resistance	Major
d. Insulation resistance	Major
e. Squib energy non-function	Major
f. Radiographic examination	Major
g. Reliability control	Major

Of these tests, (g) is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Seal integrity, bridge resistance, circuit resistance, insulation resistance, squib energy non-function, and radiographic examination. Each switch shall be tested in accordance with 4.4.3 thru 4.4.7 and 4.4.9.1. If any item fails to comply with 3.5 thru 3.8, 3.9.1 and 3.10.1, the item shall be rejected and removed from the lot.

4.3.3.2 Reliability control. Sample of 32 units from a given switch type lot shall be subjected to the tests in accordance with 4.4.10 and comply with the requirements of 3.11.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL)(see 2.1, PUBLICATIONS). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Factorial test plan. Each group of 16 units of a given switch type (see 4.2.1) shall be subjected to all tests in the sequence of treatment as indicated in Table VI, embodying a factorial screening pattern designed to assess the effects of the prescribed environments. Analysis of the factorial screening test results will, in any case, be the responsibility of the Commander, Picatinny Arsenal, ATTN: SARPA-QA-N-A, Dover, N.J. 07801

Table VI

SEQUENCE of TREATMENT

Seal integrity (see 3.5.1) & (3.5.2)-All units, variables data on 3.5.2.

Bridge resistance (see 3.6) - All units, variables data.
Circuit resistance (see 3.7) - All units, variables data.
Insulation resistance (see 3.8) - All units, variables data.

Squib energy non-function (see 3.9.1) - All units, attribute data.
Radiographic examination (see 3.10.1) - All units, attribute data.

Sample Unit Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Note (1)
A-Temperature shock (see 3.10.2)			+			+	+	+	+				+		+	+	
B1-Transportation vibration (see 3.10.3) (low temperature, 65 ± 5°F)	+	+	+			+				+			+	+		+	
B2-Transportation vibration (see 3.10.3) (high temperature, +160 ± 5°F)				+	+		+	+	+		+				+		
C-Mechanical shock (see 3.10.4)			+			+	+	+		+	+	+		+			
D-Random vibration (see 3.10.5)		+			+	+	+		+			+	+	+			
E-Low temperature (see 3.10.6)			+	+	+	+				+			+	+			+

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

Sample Unit Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
F-High temperature (see 3.10.7)			+	+	+	+			+		+			+	+		
G-Temperature and humidity (see 3.10.8)		+	+		+			+	+	+			+		+		
Radiographic examination (see 3.10.1.1) - All units, variables data.																	
Bridge resistance (see 3.6) - All units, variables data.																	
Circuit resistance (see 3.7) - All units, variables data.																	
Insulation resistance (see 3.8) - All units, variables data.																	
Squib energy non-function (see 3.9.1) - All units, attribute data.																	
H1-Low temperature function (see 3.10.6.2)	+				+	+	+					+	+	+	+		Note (2)
H2-High temperature function (see 3.10.7.2)			+	+	+			+		+	+	+				+	Note (3)
Radiographic examination, post-function (see 3.10.1.2) - All units, attribute data.																	

Note (1) Each sample unit to be subjected to that sequence of environmental treatments (A-H) indicated by plus (+) marks in the corresponding unit column.

Note (2) In H1, variables data to be obtained on function time and post-function circuit resistance in that order.

Note (3) In H2, variables data to be obtained on function time, post-function insulation resistance, and circuit resistance in that order.

4.4.2 Probability of compliance. The probability of compliance will be determined by first computing the marginal averages and pooled standard deviation of those items in the factorial experiment, Table VI, which received each particular environment. Using the marginal averages as the sample average ($\bar{X}$), the corresponding "K" - value (Normal deviate) for each environment will be obtained via difference comparison with the specified upper (UL) of lower (LL) specification limit, said difference being normalized in terms of the sample standard deviation(s).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

Thus: $K_U = \frac{UL-\bar{X}_U}{s}$ and, $K_L = \frac{\bar{X}_L - LL}{s}$

Where: K_U, K_L=Normal deviates associated with the upper and lower specification limits, respectively.

$\bar{X}_U$, $\bar{X}_L$=The average calculated from the mariginal total for the upper and lower levels, respectively, at the environment of interest.

s =The overall standard deviation.

UL, LL=The upper and lower specification limits, respectively.

4.4.3 Seal integrity.

4.4.3.1 Gross leakage. The switch shall be placed into an approved sealed test chamber and the chamber purged and backfilled with dry nitrogen to 15 ± 2 pounds per square inch (psi) gage. After pressurization for a duration of 2-1/2 to 3-1/2 hours, the switch shall be exposed to normal atmosphere and immersed in 2 inches of distilled water. Taking care to wipe trapped air bubbles free from immersed surfaces, switches shall be observed for leakage bubbles. The air over the surface of the water shall be evacuated to a pressure of 5 ± 1 (psi) absolute. Trapped air pockets may be removed by agitation or jarring during initial evacuation. While at this pressure for a minimum of 60 seconds, the switch shall be observed for evidence of air bubble leakage. Discretionary care should be exercised in eliminating immersion groupings to quantities sufficiently small to enable accurate observation. Only such quantity of switches as may be bubble checked within 2-1/2 hours of decompression shall initially be pressurized. Upon completion of test, each switch shall be brush cleaned with distilled water to remove residue traces from glass seal areas. Following cleaning, the switches shall be vacuum dried at 125 ± 5°F, and an absolute pressure of 55 millimeters (mm) of mercury maximum; for a minimum of 1 hour. After testing, the switch shall be tested as specified in 4.4.3.2.

4.4.3.2 Fine leakage. The switch shall be placed into an approved hermetically sealed test chamber and the chamber evacuated to less than 1 inch of mercury. Following an evacuation period of 4 ± 1/2 hours, the chamber shall be backfilled with oil-free, 99.5 percent pure helium gas containing not more than 0.02 milligrams (mg) of water per liter. The backfilled pressure shall be brought to 15 ± 1 psi gage and maintained for 16 ± 2 hours. Following release of the chamber pressure the switch shall be removed and placed in the hermetically sealed chamber of a fixed focus mass spectrometer

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

having a capability of measuring to 1 x 10^{-6} atmosphere cc/second. The spectrometer and chamber shall be evacuated to 0.2 microns maximum and readings observed to determine compliance with 3.5.2.

4.4.4 Bridge resistance. The electrical resistance of of squib bridge wire shall be measured through the squib terminals with an approved milliohmmeter using a four terminal method. The current shall be limited to 5 milliamperes dc maximum: or root mean square equivalent, to determine compliance with 3.6.

4.4.5 Circuit resistance. The circuit resistance of the closed switch contacts shall be measured through the switch terminal with an approved milliohmmeter using a four terminal method. The current shall be limited to 10 milliamperes maximum, to determine compliance with 3.7.

4.4.6 Insulation resistance. The insulation resistance between designated switch points shall be measured at a test potential of 500 + 50 volts dc applied for a minimum of 5 seconds and a maximum of 60 seconds using an approved megohmmeter. The maximum current output of the megohmmeter shall be 5 milliamperes. Post-function insulation measurements shall begin with the switch housing test point and end with a repeat of the same. The initial post-function reading shall be obtained within 10 seconds, maximum, of switch functioning. Test results shall comply with 3.8.

4.4.7 Squib energy non-function. The squib terminals shall be subjected to 100 + 10 - 0.00 milliamperes of dc for a minimum of 10 seconds. The position of the switch contacts shall thereafter be tested for circuit continuity to determine if the switch has functioned.

Note: Each squib switch is marked with a circuit schematic indicating the position of the switch contacts prior to functioning.

4.4.8 Squib energy function. The switch shall be subject to 1.00 + 0.00 - 0.05 ampere dc applied to the squib terminals to initiate the squib. The test for switch functioning, defined as the opening of all normally closed contacts and the closing of all normally open contacts, shall be verified after functioning by testing the switch for compliance with the post-function requirements of insulation resistance (3.8) and circuit resistance (3.7), in that order. Functioning time, which commences with first application of the above current pulse, shall be measured with approved equipment. After functioning, the switch shall be submitted to post-function radiographic examination (3.10.1.2).

4.4.9 Environmental testing (see 3.10).

4.4.9.1 Radiographic examination. All radiographic inspection shall be accomplished by using the film technique method. The switch assembly shall be X-rayed in accordance with the procedures specified in MIL-STD-271 and

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

Neutron radiographed in accordance to the best commercial procedures. The assembly shall be X-rayed in two longitudinal positions, respectively, with terminal pins up and with terminal pins on the side. The assembly shall be neutron radiographed in a longitudinal position with the terminal pins on the side. The method of film examination shall be visual and radiographic records must be arranged to maintain individual switch identities. The requirements specified in 3.10.1 must be complied with.

4.4.9.2 Temperature shock. The switch shall be tested in accordance with the procedure in MIL-E-5272, except each temperature cycle shall be at +160 + 5°F (15 percent relative humidity maximum) for 8 hours and -65 + 5°F for 16 hours. After three consecutive cycles totaling 72 hours minimum, the switch shall be removed from the test chamber and examined visually for compliance with 3.10.2.

4.4.9.3 Transportation vibration. The switch shall be located within a temperature test chamber on a suitable fixture that in rigidly mounted to a vibration machine. After the switch has been conditioned at the indicated temperature extreme (+160 + 5°F or -65 + 5°F) for a minimum of 2 hours, it shall be sweep cycled in accordance with the frequency ranges specified in 3.10.3. Cycling of the switch shall be continuous. Vibration shall be applied in each of the three mutually perpendicular axes of the switch. A 30-minute temperature restabilization period shall follow each reorientation of the switch axis prior to recycling to the vibration requirement. After the vibration has been completed and the switch returned to room temperature, an examination shall be made, visually, to determine compliance with 3.10.3.

4.4.9.4 Mechanical shock. The switch shall be placed in a suitable fixture and rigidly mounted on a shock machine. After the shock requirement has been applied in each of three principal axis the switch shall be examined, visually, to determine compliance with 3.10.4.

4.4.9.5 Random vibration. The switch shall be placed in a suitable fixture and rigidly mounted to a vibration machine. The random vibration schedule shall be in two parts and in accordance with MIL-STD-810, Method 514.1.

Part I, shall consist of the Random Vibration Test Curve Envelope, Figure 514-4, Test Curve F, except, that the time shall be reduced to 10 seconds.

Part II, shall consist of the Random Vibration Test Curve Envelope, Figure 515-4, Test Curve C, except, that the time shall be reduced to 3 minutes.

After the vibration requirements have been applied in each of the three mutually perpendicular axis the switch shall be examined, visually, to determine compliance with 3.10.5.

4.4.9.6 Low temperature.

4.4.9.6.1 Soak. The switch shall be placed in an approved temperature test chamber and exposed to -65 + 5°F for a minimum of 72 hours. After soaking for the prescribed time, the switch shall be returned to room temperature and examined, visually, to determine compliance with 3.10.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

4.4.9.6.2 Function. The switch shall be placed in an approved temperature test chamber and conditioned at -65 ± 5°F for a minimum of 2 hours. After the initial conditioning period and while still at -65°F, the switch shall be functioned and tested for compliance with 3.10.6.2.

4.4.9.7 High temperature.

4.4.9.7.1 Soak. The switch shall be placed in an approved temperature test chamber and exposed to +160 ± 5°F for a minimum of 72 hours. After soaking for the prescribed time, the switch shall be returned to room temperature and examined visually, to determine compliance with 3.10.7.1.

4.4.9.7.2 Function. The switch shall be placed in an approved temperature test chamber and conditioned at +160 ± 5°F for a minimum of 2 hours. After the initial conditioning period and while still at +160°F, the switch shall be functioned and tested for compliance with 3.10.7.2.

4.4.9.8 Temperature and humidity. The switch shall be placed in an approved test chamber and tested in accordance with MIL-STD-331, Test No. 105: except, the test shall consist of one complete 14-day "JAN TEMPERATURE AND HUMIDITY CYCLE" only, and the temperature extreme shall not exceed +160 ± 5°F or -65 ± 5°F. After test the switch shall be removed from the test chamber and examined, visually, for compliance with 3.10.8. All exterior surfaces of the switch shall be cleaned with technical grade ethanol and a camel hair brush to remove dirt or other residue. Note: Do not remove lacquer from marked surfaces with ethanol.

4.4.10 Reliability control. A sample of 32 switches shall be selected at random in such manner as to represent each portion of each lot. The sample shall be retested to tests a, b, c, d and e in the order indicated in 4.3.3, and variables data recorded.

Group I. 16 switches shall be placed in an approved temperature test chamber and conditioned at +195 ± 5°F, each switch shall be functioned as specified in 3.9.2 and tested for compliance with the high temperature requirement 3.11.

Group II. 16 switches shall be placed in an approved temperature test chamber and conditioned at -80 ± 5°F for a minimum of 2 hours. After the initial conditioning period and while still at -80 ± 5°F, each switch shall be functioned as specified in 3.9.2 and tested for compliance with the low temperature requirement 3.11.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The switches shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. The switches shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The switches shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Physical damage. Physical damage will include cracks in glass seal, bent pins, cracks in pin, cracks in housing, distortion of housing.

6.6 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2028 dated 11 May 1971, with Amendment 3, dated 23 March 1976.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A133

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-N-Q
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:21Z
Check the source to verify that this is the current version before use.